

EE4730TU High Frequency Wireless Architectures

Project: Material Characterization using a Time Domain System

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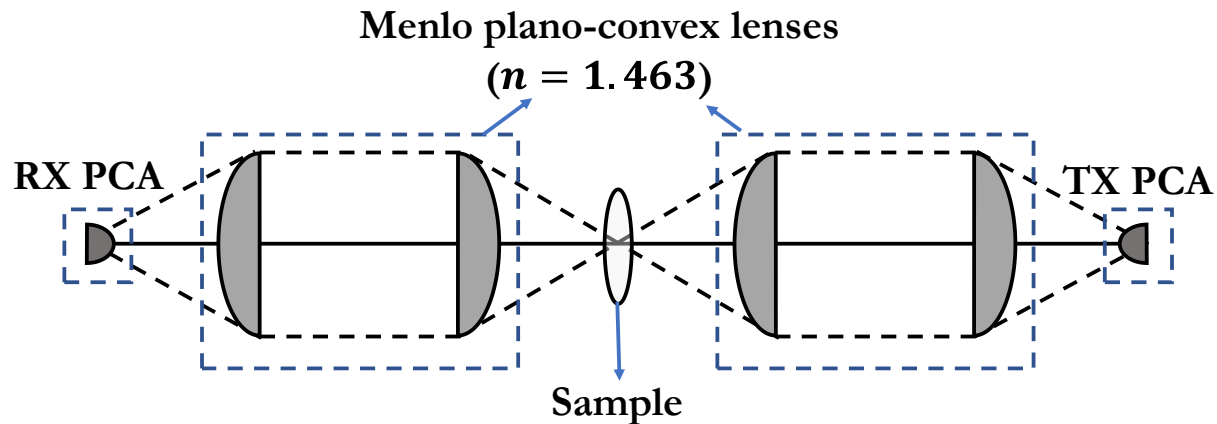
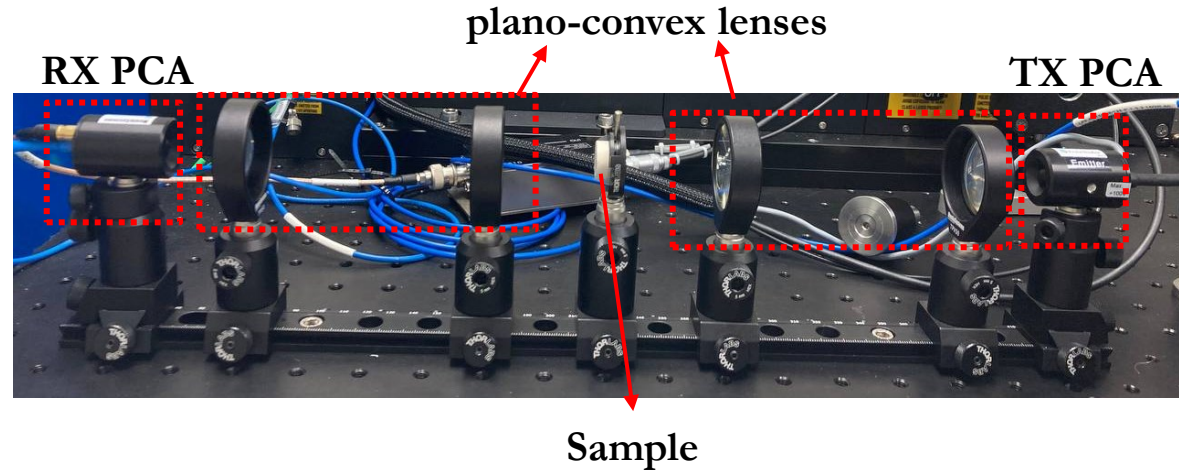
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Material Characterization with a TD System

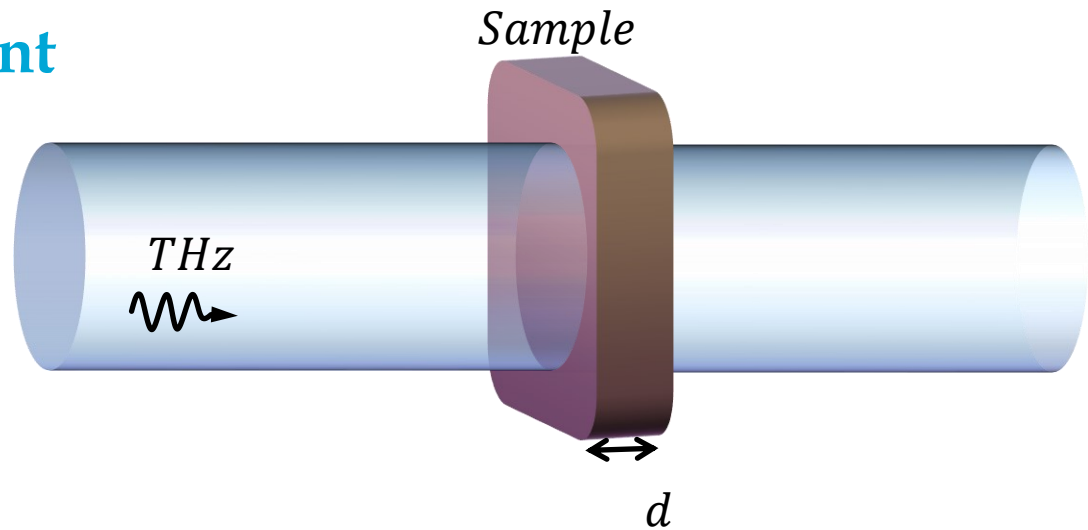
Menlo Tera K15



Objective:

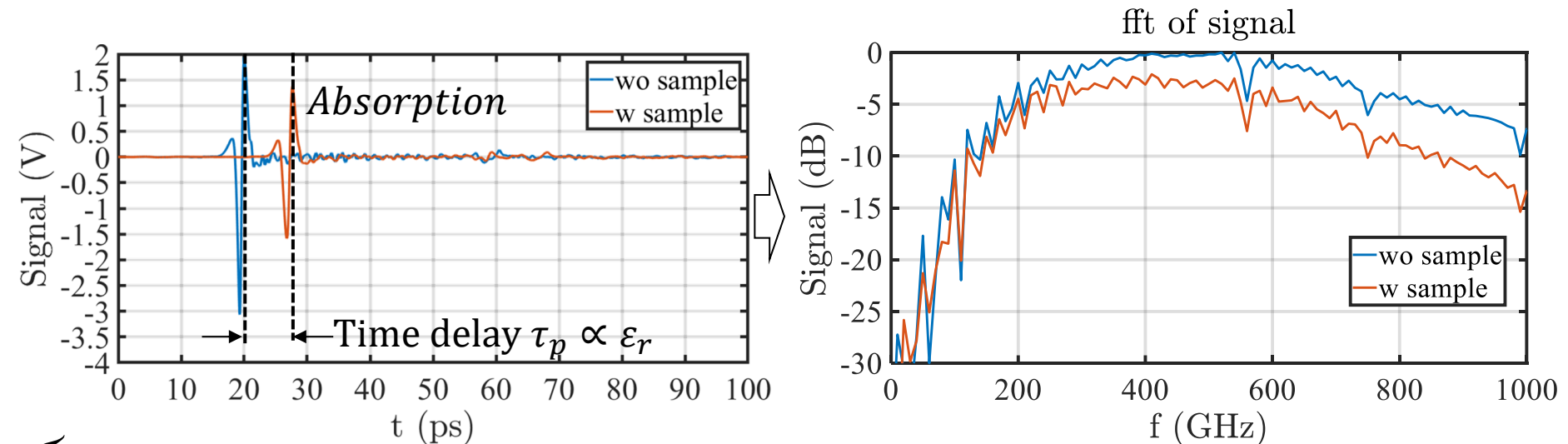
- Understand the manipulation of TD equipment and calibration procedures
- Analyse the material characterization capabilities of the TD system.
- Apply transmission line models to extract the permittivity ϵ_r , the attenuation in dB/lambda and loss tangent $\tan\delta$ of different materials:
 - Silicon, Quartz, PREMIX...

Measurement

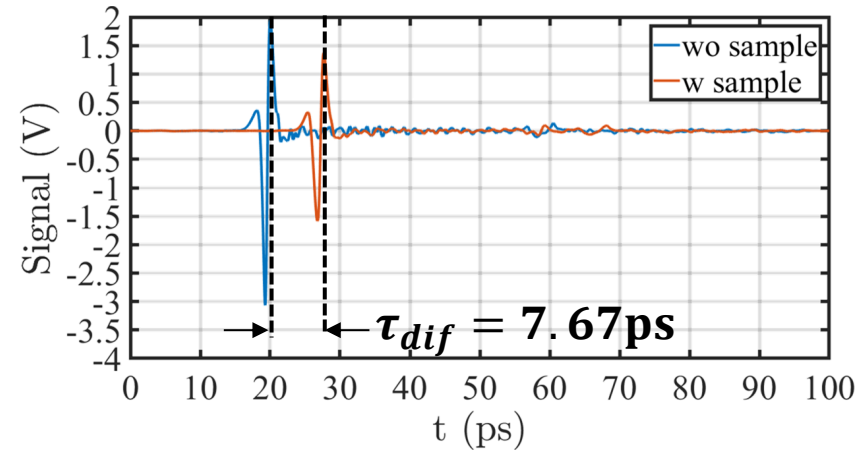
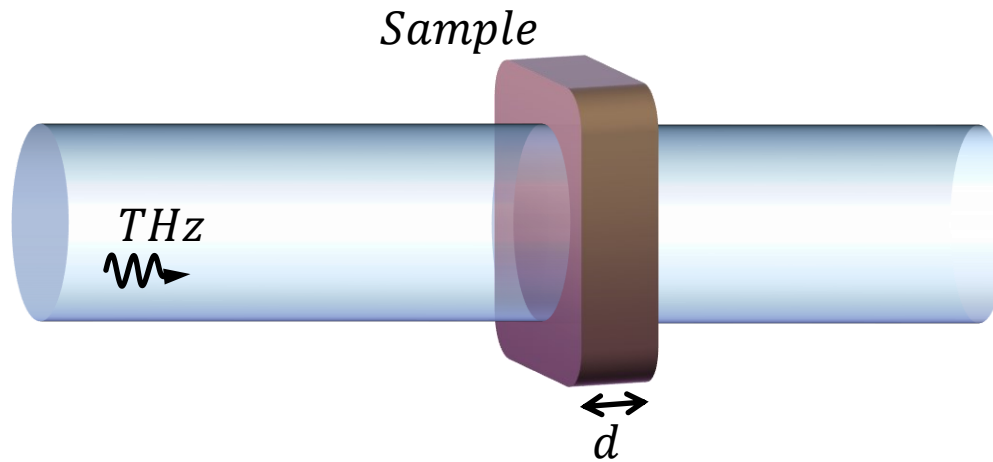


In the laboratory you will perform a differential measurement:

- 1. Measure the time response without sample
- 2. Measure the time response with the sample



Obtaining the permittivity



The permittivity can be extracted from the time delay between the two pulse measurements:

$$\tau_{dif} = \tau_d - \tau_0$$

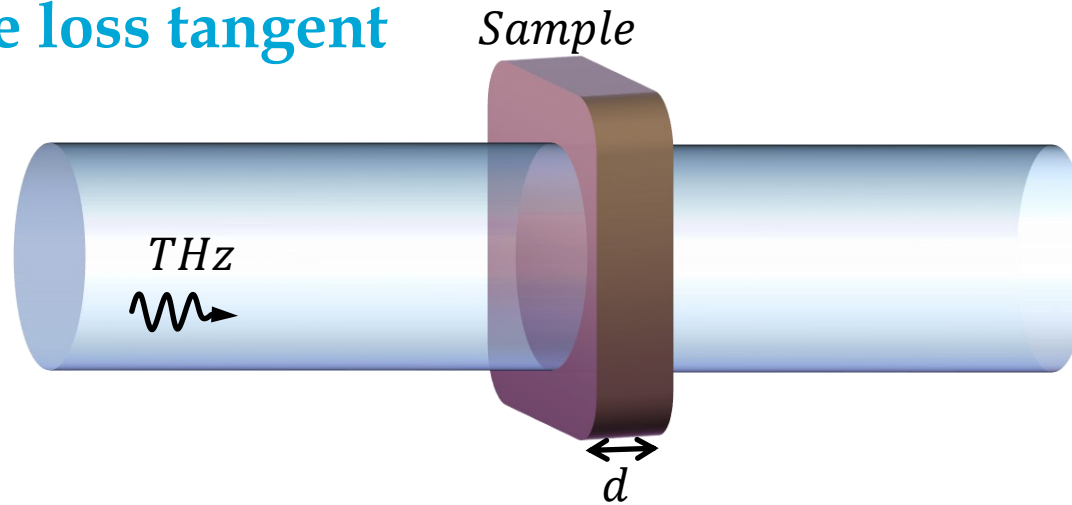
τ_d : time it takes for the pulse to go through the sample

τ_0 : time it takes for the pulse to go through air over the thickness of the sample

Knowing that $\tau_0 = \frac{d}{c_0}$ and $\tau_d = \frac{d\sqrt{\epsilon_r}}{c_0}$

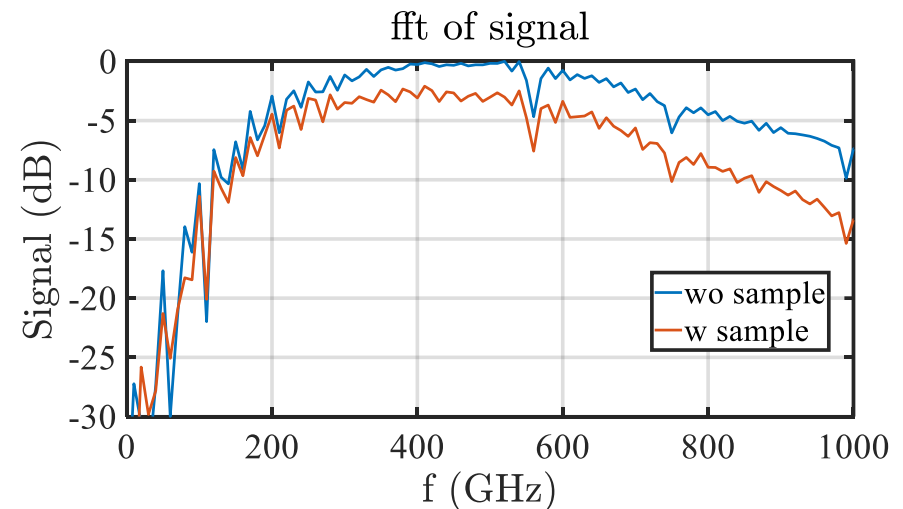
$$\epsilon_r = \left(\frac{c_0 \tau_d}{d} \right)^2 = \left(\frac{c_0 (\tau_{dif} + \tau_0)}{d} \right)^2$$

Obtaining the loss tangent

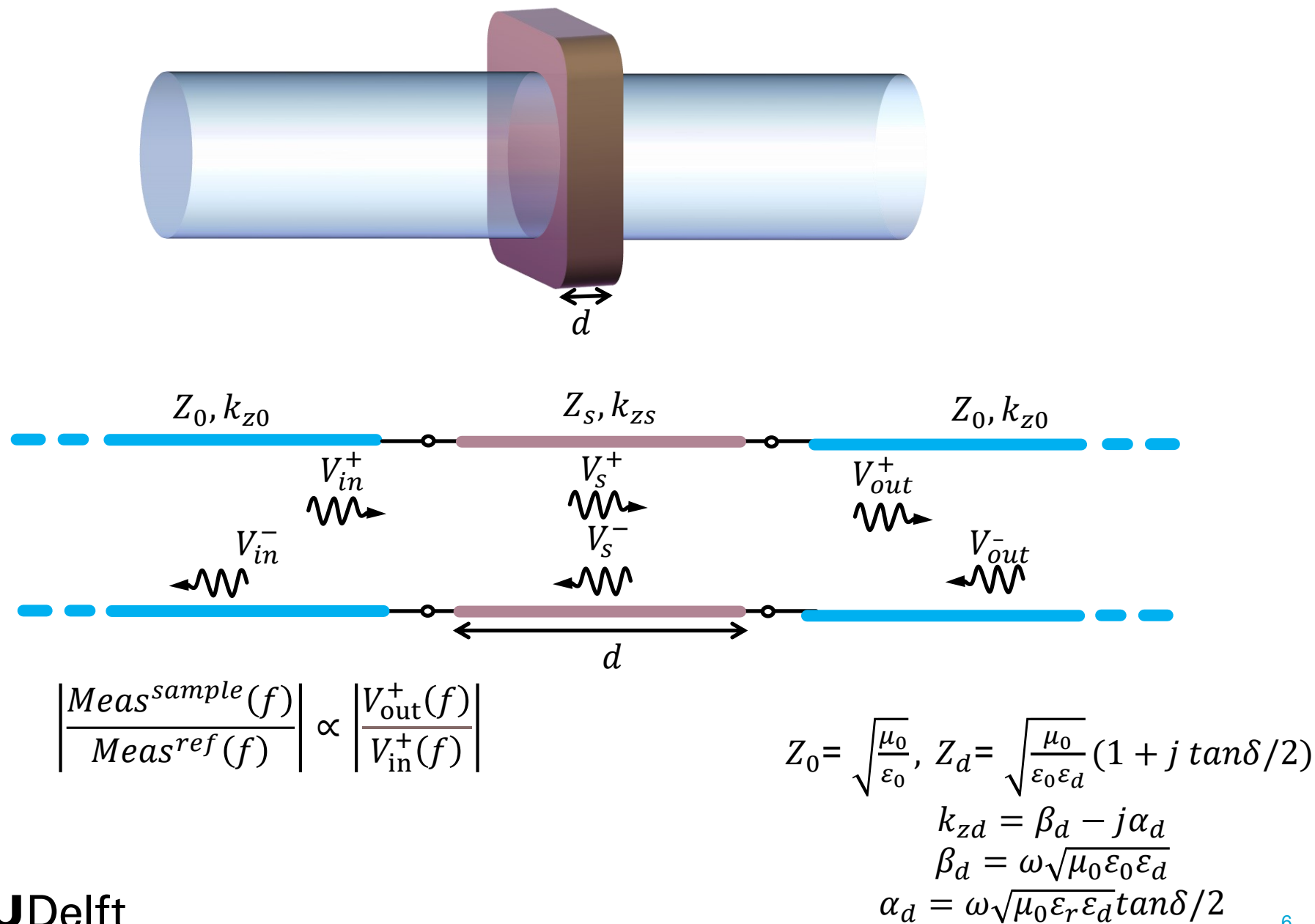


Processing:

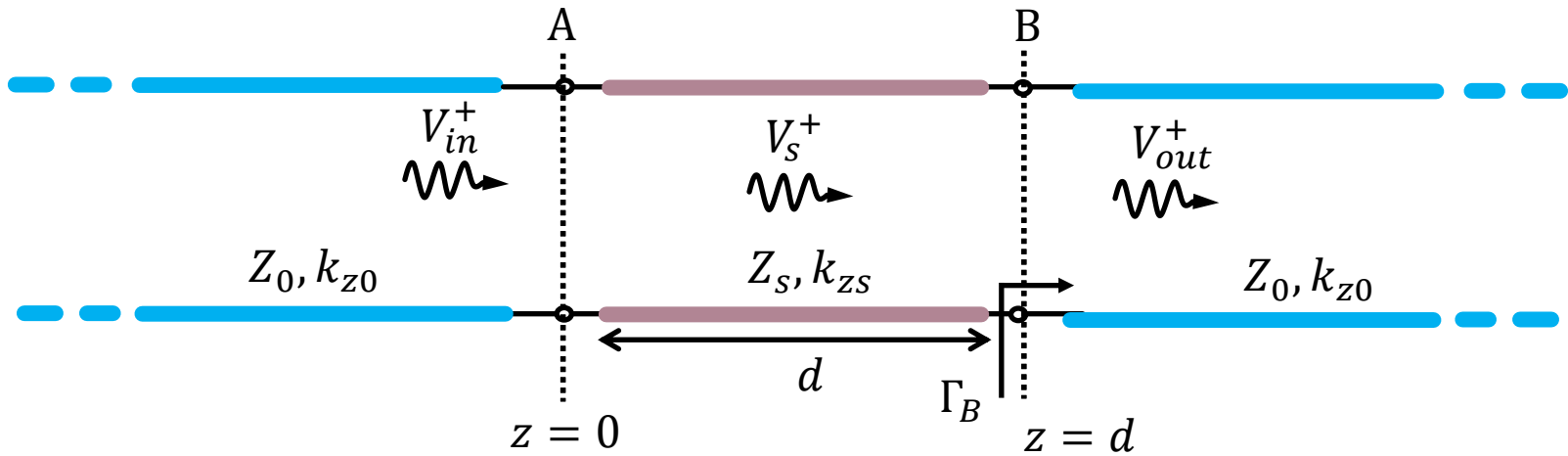
1. Convert your signals from the time domain to the frequency domain $Meas^{sample}, Meas^{ref}(f)$
2. Obtained the difference between the two responses
3. Find the permittivity and loss tangent in you transmission line model that fit the measured data



Transmission line model



Transmission line model derivation



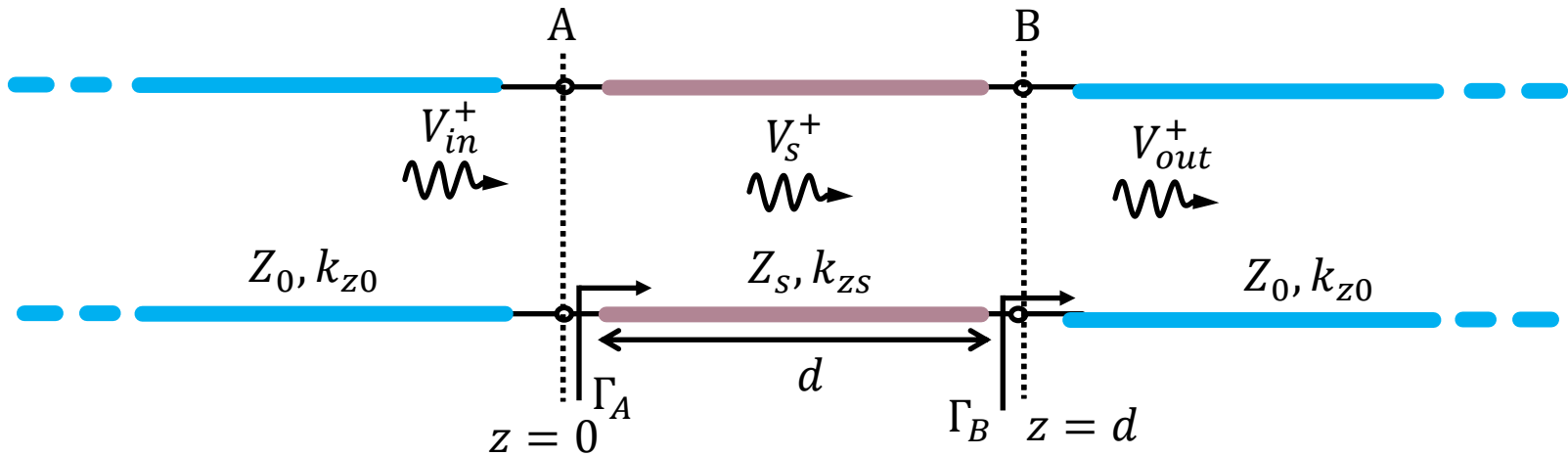
$$V_S(z) = V_S^+ e^{-jk_{zs}z} + V_S^- e^{jk_{zs}z} = V_S^+ [e^{-jk_{zs}z} + \Gamma_B e^{jk_{zs}(z-2d)}]$$

$$V_{out}(z) = V_{out}^+ e^{-jk_{z0}z} \quad \Gamma_B = \frac{V_S^- e^{jk_{zs}d}}{V_S^+ e^{-jk_{zs}d}} = \frac{Z_0 - Z_S}{Z_0 + Z_S}$$

At boundary B: $V_{out}(z = d) = V_S(z = d)$

$$\frac{V_{out}(z = d)}{V_S(z = d)} = \frac{V_{out}^+ e^{-jk_{z0}d}}{V_S^+ e^{-jk_{zs}d} [1 + \Gamma_B]} = 1 \quad \rightarrow \quad \frac{V_S^+}{V_{out}^+} = \frac{e^{jk_{zs}d} e^{-jk_{z0}d}}{1 + \Gamma_B}$$

Transmission line model derivation



$$V_s(z) = V_s^+ e^{-jk_{zs}z} + V_s^- e^{jk_{zs}z} = V_s^+ [e^{-jk_{zs}z} + \Gamma_B e^{jk_{zs}(z-2d)}]$$

$$\Gamma_A = \frac{Z_{in}^A - Z_0}{Z_{in}^A + Z_0}$$

$$V_{in}(z) = V_{in}^+ e^{-jk_{z0}z} [1 + \Gamma_A]$$

$$Z_{in}^A = Z_s \frac{Z_0 + jZ_s \tan(k_{zs}d)}{Z_s + jZ_0 \tan(k_{zs}d)}$$

At boundary A: $V_{in}(z=0) = V_s(z=0)$

$$\frac{V_{in}(z=0)}{V_s(z=0)} = \frac{V_{in}^+ [1 + \Gamma_A]}{V_s^+ [1 + \Gamma_B e^{-jk_{zs}2d}]} = 1$$

$$\frac{V_{in}^+}{V_s^+} = \frac{[1 + \Gamma_B e^{-jk_{zs}2d}]}{[1 + \Gamma_A]}$$

$$\frac{V_{in}^+}{V_{out}^+} = \frac{V_{in}^+}{V_s^+} \frac{V_s^+}{V_{out}^+} = \frac{e^{jk_{zs}d} e^{-jk_{z0}d} [1 + \Gamma_B e^{-jk_{zs}2d}]}{[1 + \Gamma_B] [1 + \Gamma_A]}$$

Comments

- File from MENLO setup: The absolute time does not start at 0. You will need to bring it to 0
- We will assume the permittivity constant in frequency. The loss tangent is frequency dependent.
- Take a look at the error when fitting the parameters.
- For further optimization you can sweep the permittivity to best fit the parameters